

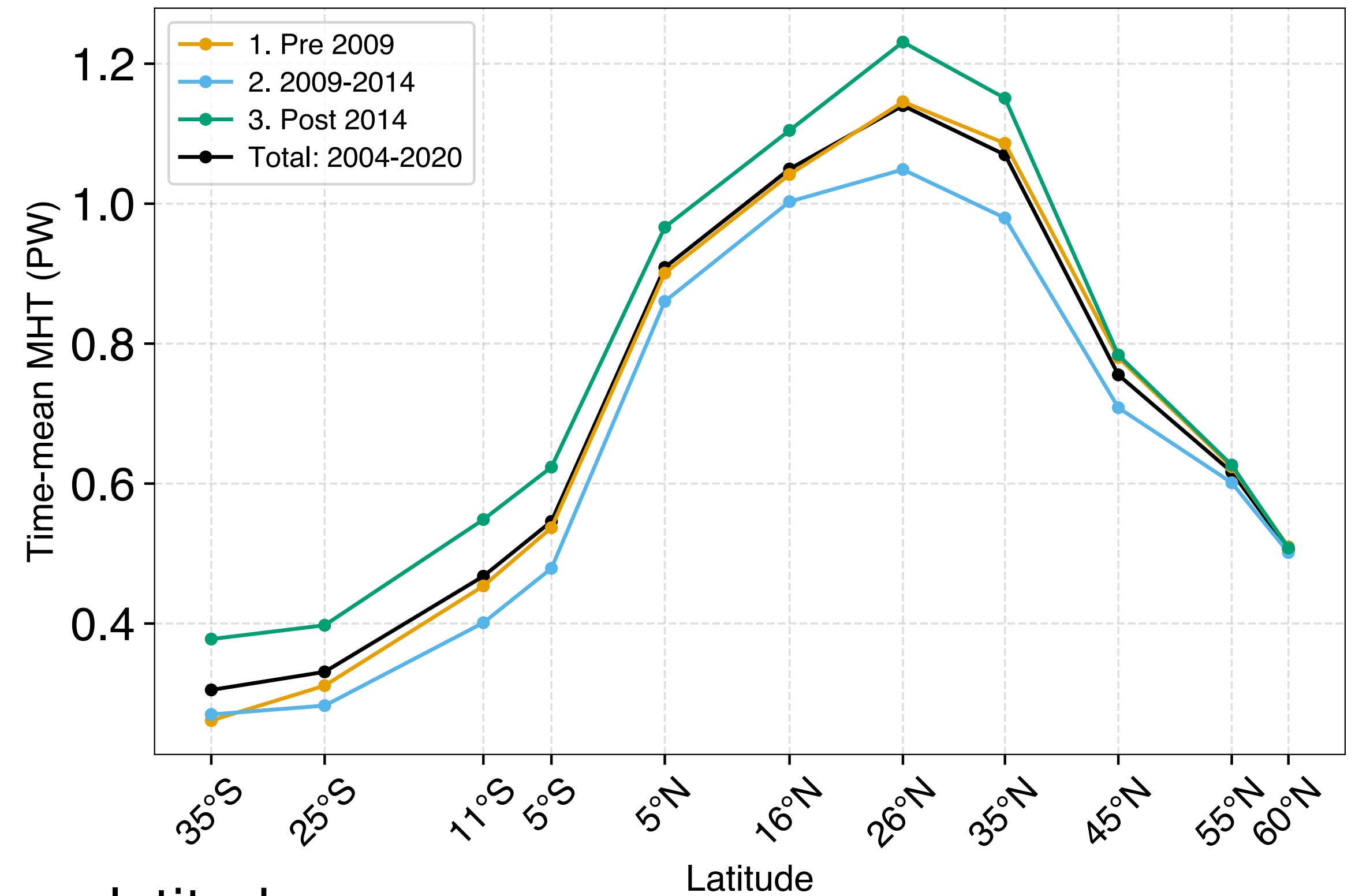
Spatiotemporal coherence of MHT in the Atlantic Ocean

4. Closing the heat budget at 16°-5°N, connecting with NAO index

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Time-mean MHT for 3 periods

Period	16°N MHT (PW)	5°N MHT (PW)
Pre 2009	1.04	0.90
2009-2014	1.00	0.86
Post 2014	1.10	0.97

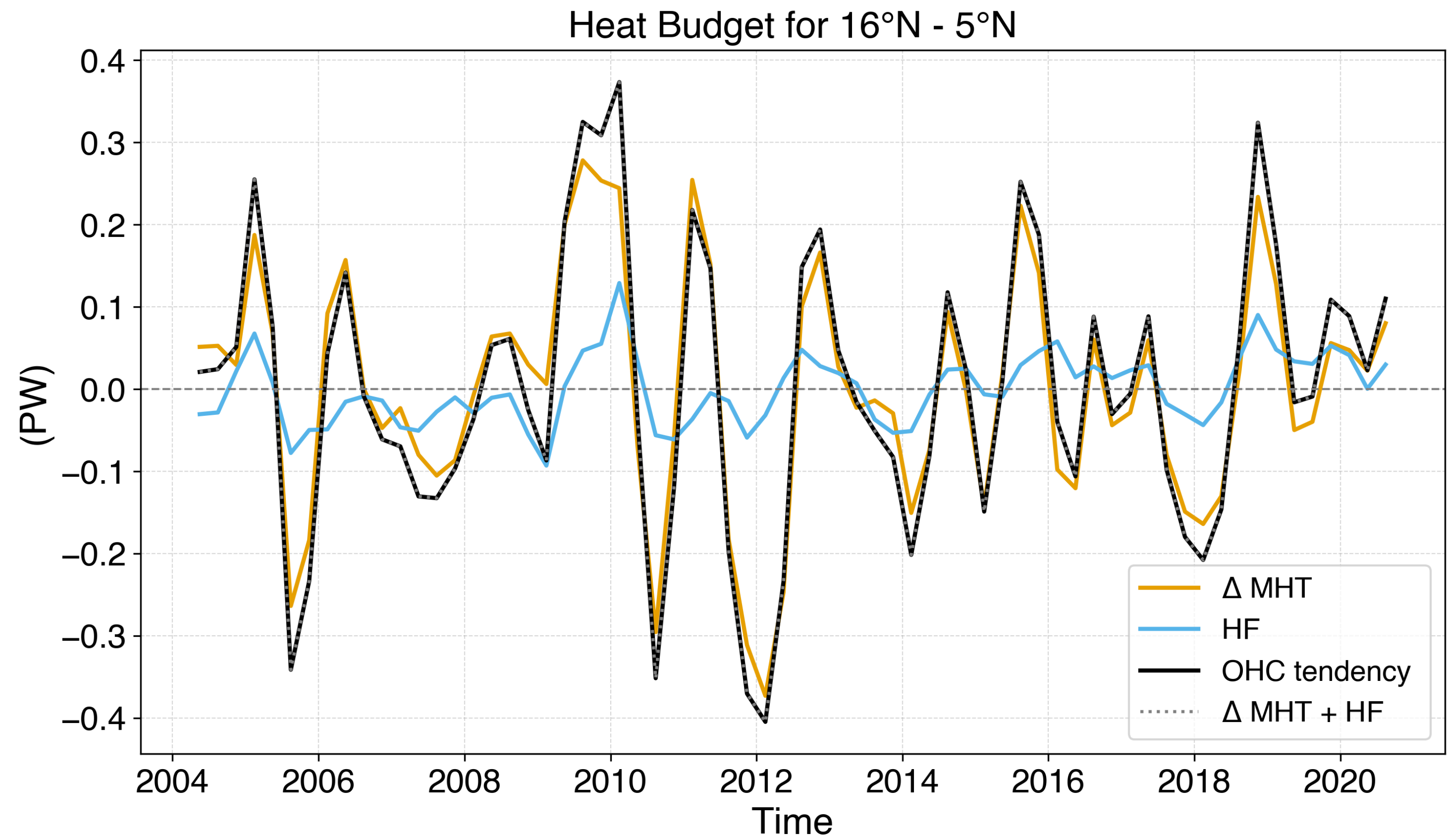


- Time mean MHT is lower in 2009-2014 period across latitudes
- Post 2014 is higher than state before coherence break
- Changes by similar amount at the two break latitudes

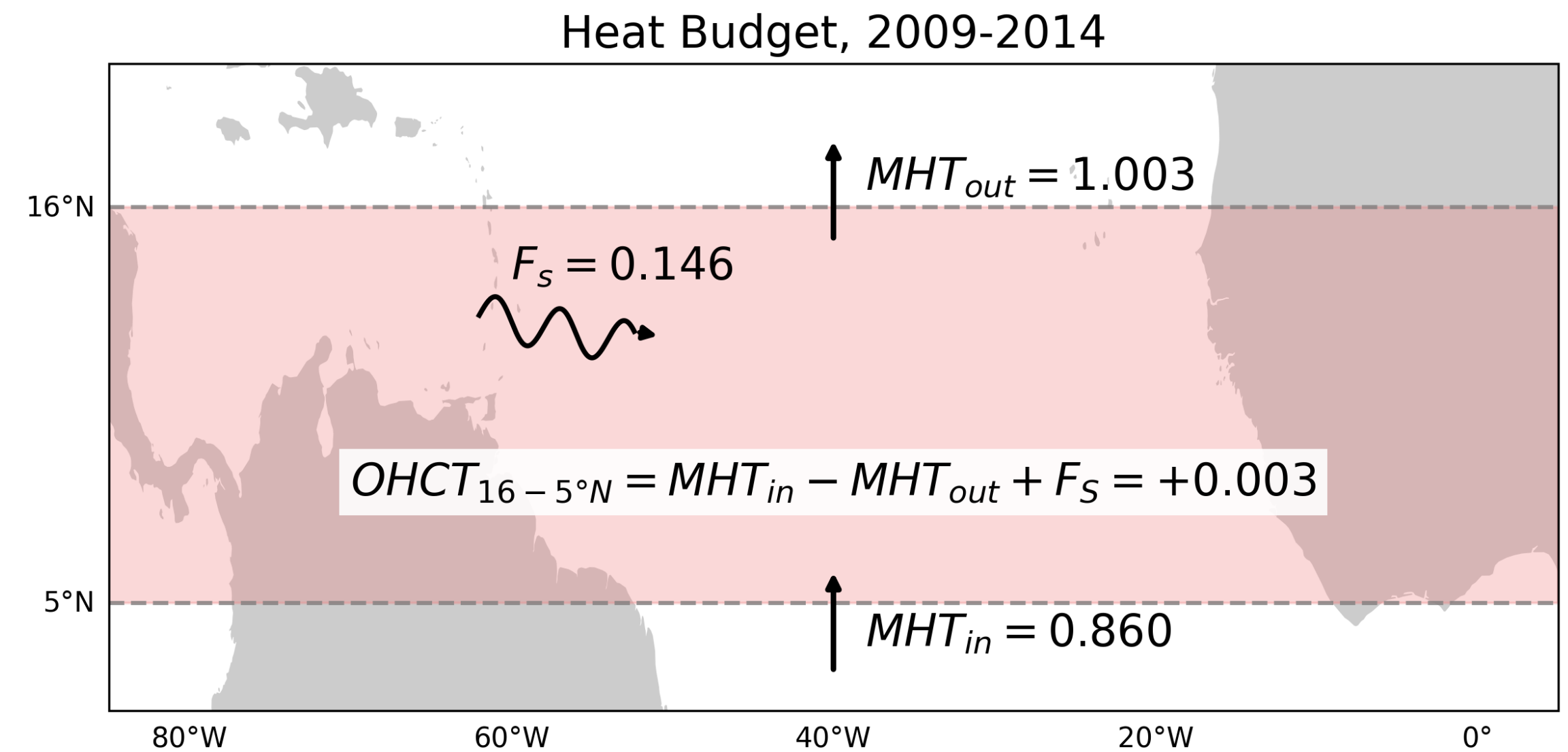
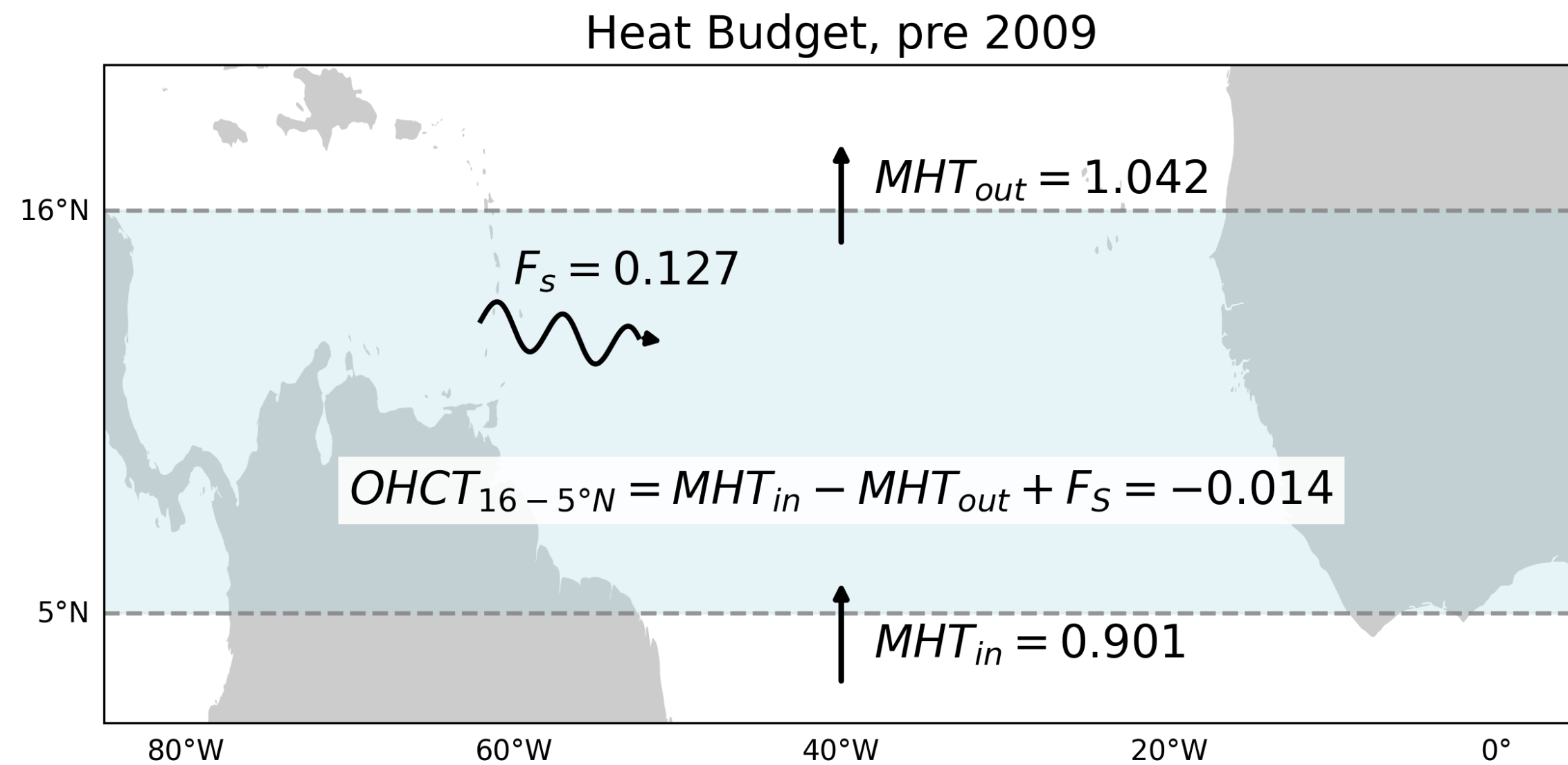
Heat budget

- How can I explain the coherence break happening in 2009-2014 at 16°- 5°N?
- How much of the change in heat is explained through air sea heat flux changes?

$$\Delta OHC/dt = \Delta MHT + \Delta F_s$$



Heat budget



2009-2014 - pre2009:

$$\Delta OHCT = +0.017 \text{ PW}$$

$$\Delta HF = +0.019 \text{ PW} \rightarrow \text{Heat flux uptake is 15\% higher in 2nd period}$$

$$\Delta HTC = -0.002 \text{ PW}$$

$\Rightarrow$ increase in OHCT is **only caused by increased uptake of heat** from atmosphere, not by a change in transport convergence inside the region

Reduction compared to pre 2009 baseline

16°N

Pre 2009:

1.042 PW

2009-2014:

1.003 PW

2009-2014 - baseline(pre2009):

-3.7% reduced MHT

5°N

Pre 2009:

0.901 PW

2009-2014:

0.860 PW

2009-2014 - baseline(pre2009):

-4.5 % reduced MHT

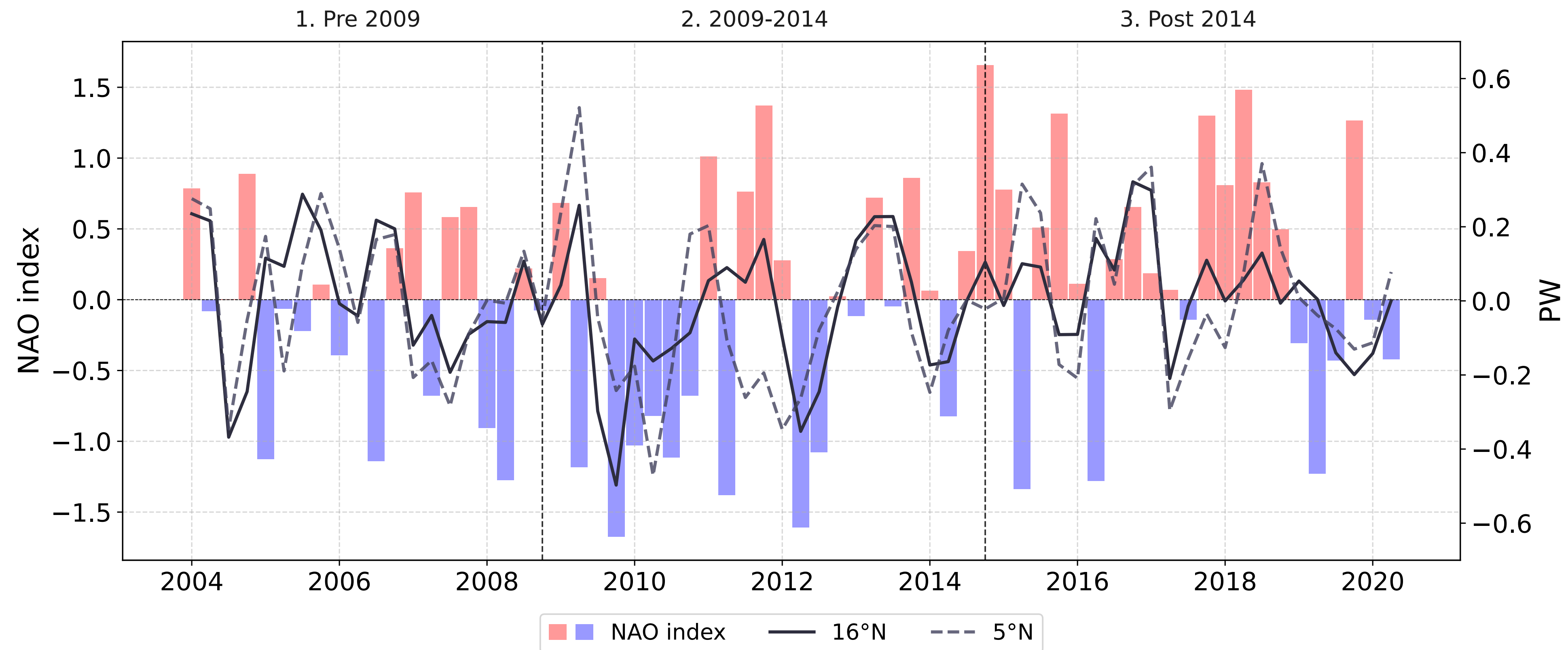
⇒ **circulation related reduction in MHT** across region ⇒ broader circulation change

However decline is almost uniform at 5°N and 16°N, which connects with almost no difference in HTC inside the region.

region leading to increase in OHCT

NAO & MHT anomalies

How do different latitudes (specifically the break latitudes) connect to the NAO index, for the different periods?



MHT and NAO correlation

- For period of negative NAO starting with the anomalous event in 2009 the heat transport in the subtropical gyre shows a higher correlation with NAO index (~0.5)
- Not visible for the other 2 periods

⇒ In period of lower over MHT, MHT anomalies in ST gyre region connect slightly with NAO index ?

Period	35°N	26°N	16°N	5°N
Pre 2009	-0.18	-0.31	-0.20	-0.17
2009-2014	0.52	0.53	0.51	0.16
Post 2014	0.21	0.12	-0.01	-0.15

Correlation of MHT at latitudes with NAO index for each period, blue shaded values are significant (95%)

Heat budget

- From literature it is known that in the time period of negative NAO event (2009/2010) there is a reduction of 2.5 Sv in AMOC from 2009-2016 (*Bryden et al. 2020*)
- State of reduced MHT across latitudes is probably caused by this decrease of overturning strength
- In such a state of reduced overturning, there is said to be more recirculation (*McCarthy 2012*)
 - heat is rather recirculated within the gyres
 - This connects to the coherence splitting between 16°N and 5°N since the subtropical gyres southern boundary is roughly in this region
 - Change in MHT is likely coming from circulation, since there is even an increase in heat uptake visible in the region 16°N-5°N
 - Correlation with NAO in this period hints towards circulation driven as well, as the „faster“ impact would be through wind & Ekman circulation changes

Gyre link

- „NAO influences Atlantic Ocean by inducing substantial changes in surface wind patterns, thereby altering the heat and freshwater exchange at the ocean surface, and influencing the thermohaline and horizontal gyres circulations (GC)“ (*NORTH ATLANTIC OSCILLATION – CONCEPTS AND STUDIES* HEINZ WANNER^{1,2}, STEFAN BRÖNNIMANN^{1,3})
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